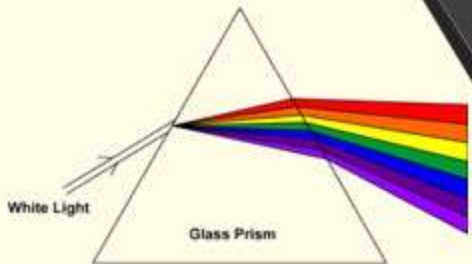


PHYSICS

MOTION AND TIME

MODULE - 2





CONSIDER THE FOLLOWING SITUATION

1

Time = 0 sec



At starting point
Distance = 0 km

2

Time = 30 sec



At destination point
Distance = 5 km





WHAT CHANGES DO YOU FIND ?

The car moves a certain distance



Time passes from 0 to 30 sec



Other objects such as the tree, the roads remains stationary.



The position of an object can either change or remain stationary with respect to time.





CONSIDER THE FOLLOWING SITUATION

Time = 0 sec



Time = 30 sec



If the position of an object does not change as time passes, it is said to be at rest



CHANGE IN POSITION w.r.t TIME... MOTION

MOTION-DEFINITION

When an object changes its position with respect to time and with respect to other stationary objects, the object is said to be in motion.

EXAMPLES



A CAR MOVING ON ROAD



A MERRY- GO- ROUND



PLANETS MOVING
AROUND THE SUN





TYPES OF MOTION



1 RECTILINEAR MOTION



ANY OBJECT MOVING
ON A STRAIGHT PATH

2 CIRCULAR MOTION



ANY CIRCULAR MOVING
OBJECT

3 PERIODIC MOTION



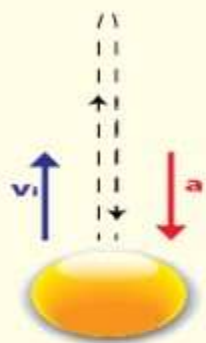
ANY OBJECT COMING BACK
TO THE ORIGINAL POSITION





1

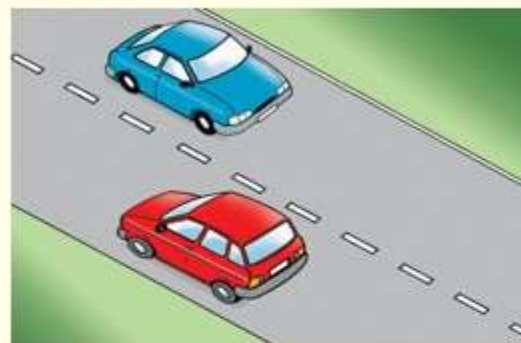
RECTILINEAR MOTION IN DAILY LIFE



**FREE FALL OF
ANY OBJECT**



**WATER FLOWING
IN A TAP**



**CARS MOVING IN
STRAIGHT LINE**



**MAN PUSHING
A LOAD**



**JOGGING IN A
STRAIGHT ROAD**



**A TRAIN MOVING IN
STRAIGHT LINES**



**SOLDIERS MARCHING
IN A PARADE**



AEROPLANE FLYING





2

CIRCULAR MOTION IN DAILY LIFE



GIANT WHEEL



CYCLE WHEEL



ELECTRONS IN ORBIT



**MOTION IN CITY
CIRCULAR FLYOVERS**



HANDS OF A WATCH



**MOTION OF
PLANETS**



**SPINNING ANY
OBJECT**



**A SPINNING MERRY
GO ROUND**



TYPES OF MOTION IN OUR DAILY LIFE ...

1 RECTILINEAR MOTION



**ANY OBJECT MOVING ON A
STRAIGHT PATH**

2 CIRCULAR MOTION



**ANY CIRCULAR MOVING
OBJECT**

3 PERIODIC MOTION



**ANY OBJECT COMING BACK
TO THE ORIGINAL POSITION**



CAN YOU SPOT THE DIFFERENCE?

A CHILD ON A SWING



PERIODIC
MOTION

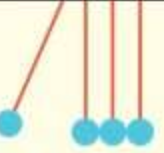
SWAYING OF A TREE



NON-PERIODIC
MOTION

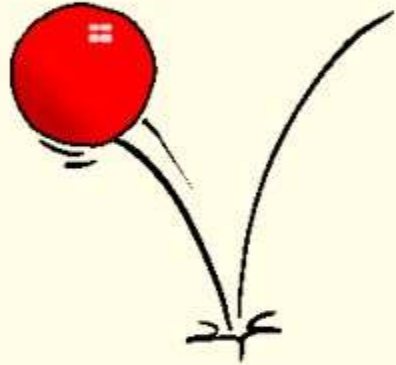
One motion is repeated after a certain interval of time and another is not bound to repeat after a certain interval





3

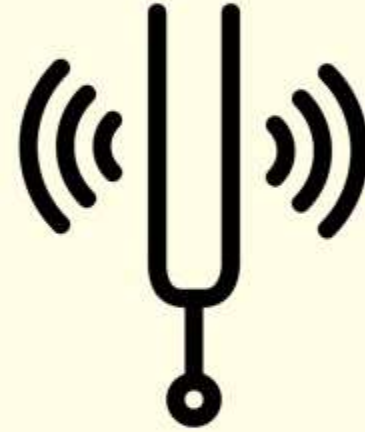
PERIODIC MOTION IN DAILY LIFE



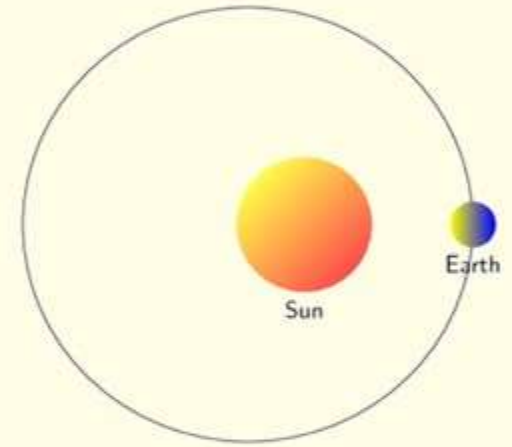
BOUNCING BALL



ROCKING CHAIR



**VIBRATING
TUNING FORK**



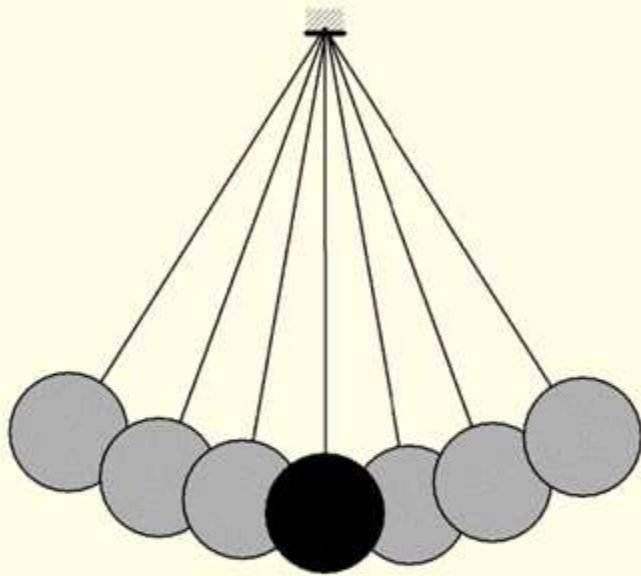
**EARTH IN ITS
ORBIT**





PERIODIC MOTION - TYPES

1 OSCILLATORY MOTION



Body moves in a pattern about its mean position.

2 VIBRATORY MOTION



Body moves about mean position due to multiple oscillations.





PERIODIC MOTION - TYPES



The 'Wobbly' London Millennium Footbridge

OSCILLATORY MOTION

- Oscillatory motion is the repeated motion in which an object repeats the same movement over and over.
- Oscillatory motion can be also found in the atoms and molecules.
- Buildings and bridges undergo oscillatory motion, sometimes that can cause disastrous effects !
- One special type of oscillatory motion is simple harmonic motion.

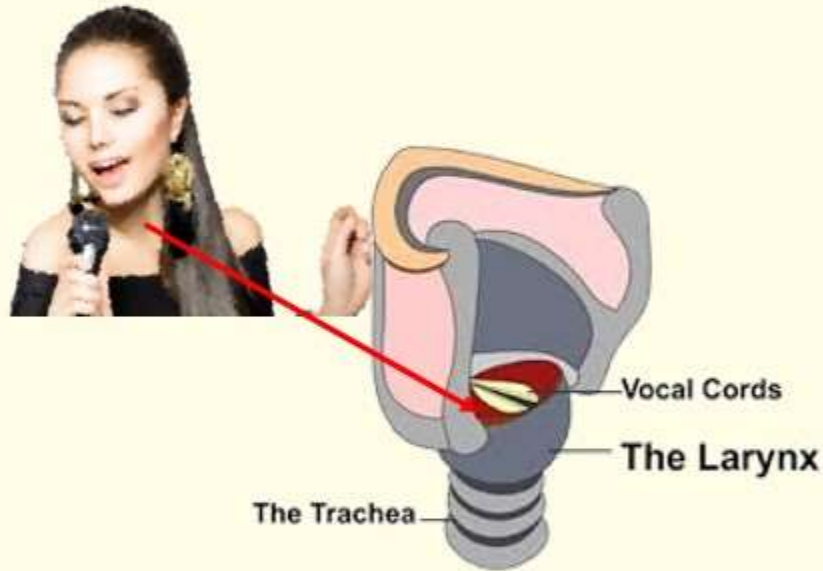
London Millennium Bridge over Thames river was immediately closed after opening in 2000 because the pedestrians witnessed severe oscillations in the bridge !





PERIODIC MOTION - TYPES

VIBRATORY MOTION



- The term vibration is used when **oscillation** have high frequency means there are multiple **oscillations** in a short time.
- The object in vibratory motion moves to and fro about a fixed point.
- We can **talk** due to Vibratory motion !
- Vibratory motion finds application in producing sound in many musical instruments like Guitar, drums, cymbals etc.

The lungs push air through the tiny slits of our vocal cords present in the larynx. These vibrate and produce sound !



LUNAR PHASES TO EXPLAIN TIME – PERIODIC ?

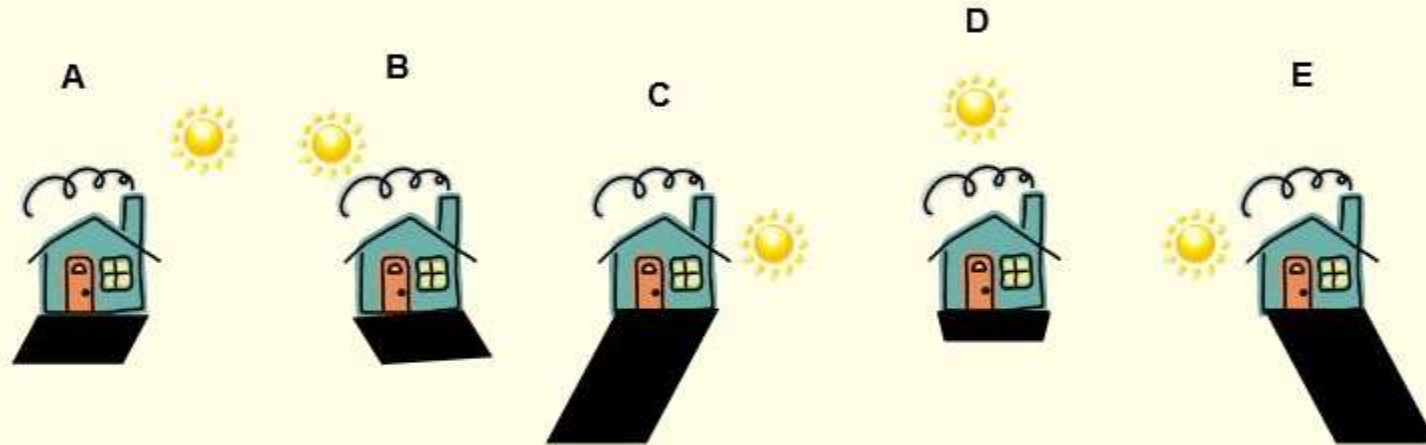


- Our ancestors noticed that many events in nature are periodic !
- We can predict a fortnight(14 days) by looking at the phases of the moon
- Similarly a month was measured from one new moon to the other.





SUN'S POSITION TO EXPLAIN TIME – PERIODIC ?



- Even if you do not have a watch, you can still predict the time of the day !
- This is by looking at the difference in the shadows at different points throughout a day.
- **Longer shadows** occur at morning and evening and **shorter shadows** occur at noon time





PERIODIC MOTION... CAN IT EXPLAIN TIME ???



JANTAR MANTAR IN
DELHI



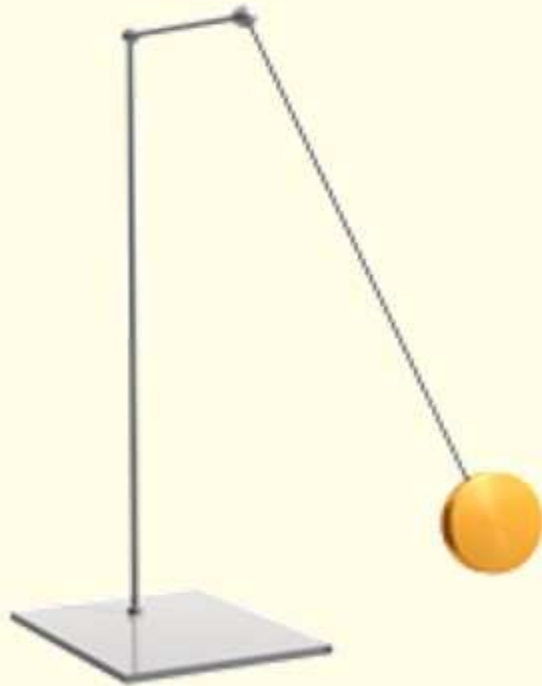
A TYPICAL
SUNDIAL

- The concept of time was discovered by looking at periodic events.
- When we have no electrical instrument to show time we have used the position of the sun and the length of the shadows to approximate the time of day it is.
- This kind of primitive clocks can be seen in Jantar-Mantar in Jaipur.
- Periodic Motion is used in clocks, helps us understand the concept of time!





PERIODIC MOTION OF A CLOCK SIMPLE PENDULUM

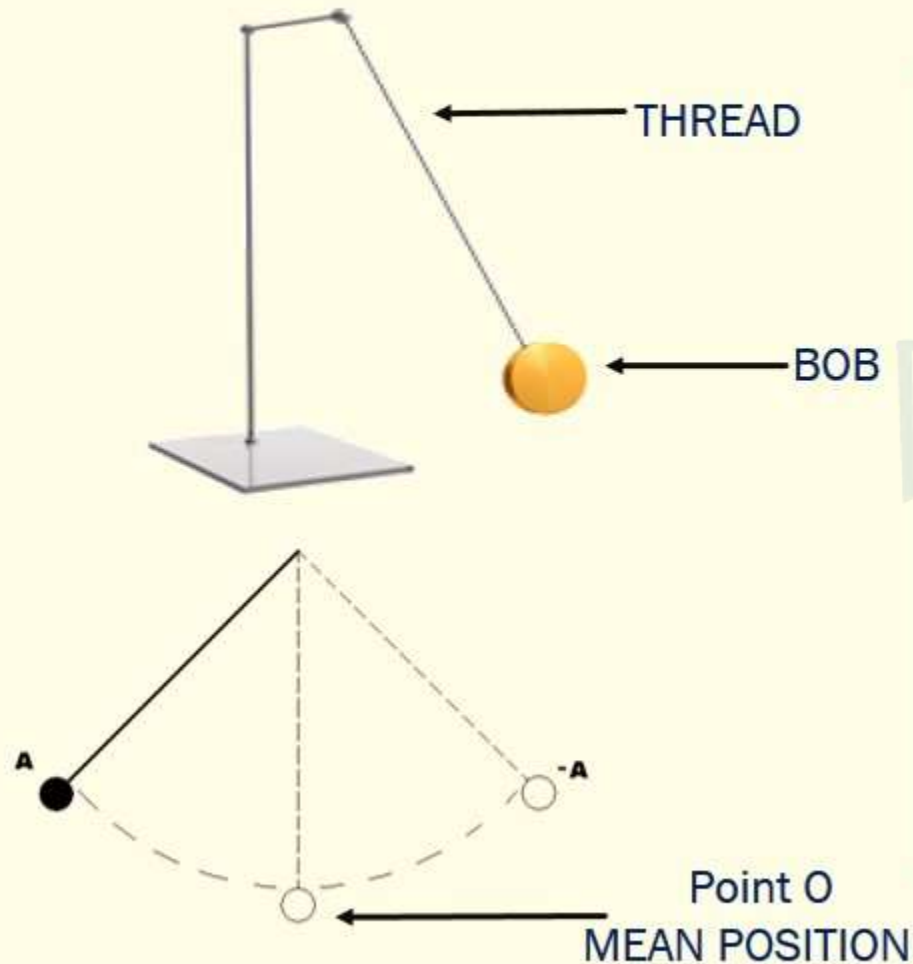


- Clocks may have very complex working mechanism but the basic principle lies in Periodic Motion.
- A very common example of periodic motion is of **Simple Pendulum** which exhibits an oscillatory motion.
- Pendulum clocks are also called **Mechanical Clocks**





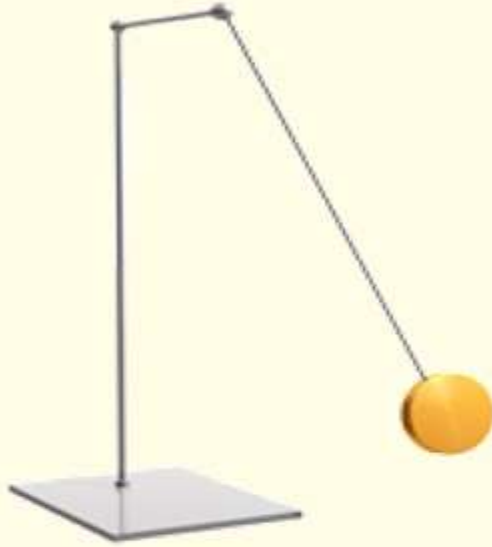
A SIMPLE PENDULUM



- When the bob of the pendulum is released from one side, it begins to move to and fro.
- This to and fro motion about a mean position is called oscillatory motion.
- The pendulum is said to have completed **one oscillation** when the bob starting from the mean position O moves to one extreme position A and again to another extreme position A' from O.



TIME PERIOD OF A SIMPLE PENDULUM



- If a stopwatch is used to measure the time taken by the bob of the pendulum to complete its oscillation from one end to another we can find the TIME PERIOD
- TIME PERIOD is defined as the time taken to complete one oscillation.
- It may be different for different pendulums !

Do you know...Because of different time periods, pendulum clocks had been replaced with quartz clock





THE CONCEPT OF QUARTZ CLOCKS



- Chemically quartz is a compound called **silicon dioxide**.
- The quartz clock uses an electronic oscillator that is regulated by quartz crystal
- Quartz is a **piezo - electric** crystal. This means they will generate electricity if they are pressed or tapped.
- They do not expand or contract when the temperature changes thus they can be used in all seasons





ATOMIC CLOCKS THE MOST PRECISE EVER!



*1 sec is precisely
9,192,631,770 cycles of
radiations of Cs atom*

- Atomic Clocks are the most precise clocks with a lag of 1 sec in over 100 million years!
- Atomic clock uses the electromagnetic waves emitted by electrons in an atom when they move to different energy levels.
- They are regulated usually by **vibrations of Cesium atoms**.
- If Cs atom is heated & the electrons goes from ground state energy level to excited state, it produces **9,192,631,770** cycles of **radiations**. These many radiations will pass through the detector in '**one**' second.





WHAT HAPPENS WHEN WE DIVIDE TIME INTO SMALLER INTERVALS ?



00:00:01



10000000000

- One second can be broken down into still smaller units and they are measureable!
- Some clocks that can measure time intervals as small as one millionth or one billionth of a second.
- One millionth of a second is called a microsecond
- One billionth of a second is called a nanosecond

Light travels exactly 11.8 inches i.e. 1 foot in 1 nanosecond. This is known as one light foot.





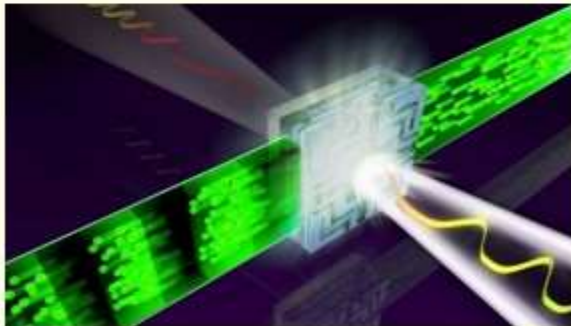
STILL SMALLER INTERVALS OF TIME ...

PICOSECOND

Picosecond is one trillionth of a second or 10^{-12} s



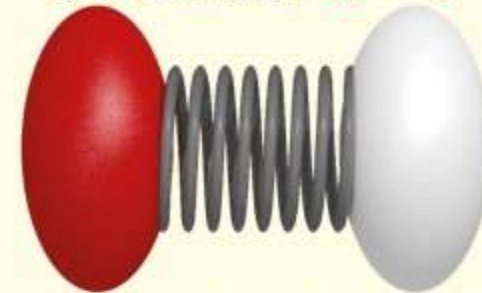
Time taken for electromagnetism to separate from other fundamental forces after BIG BANG.



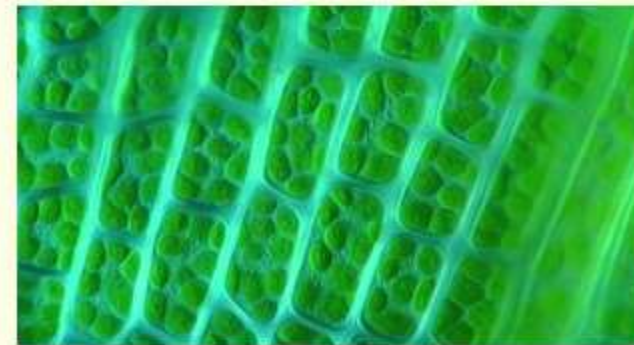
Time taken for light to travel in millimeter distance.

FEMTOSECOND

Femtosecond is one quadrillionth of a second or 10^{-15} s



Time taken for the vibration of atoms



Useful in measuring the swiftest chemical reaction of photosynthesis



WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

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G. V. Rao

Amrita Naidu

D. Sunil Kumar

L. V. Krishna

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